

JEE MAIN REVISION MASTER: HYDROCARBONS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 15 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Hydrocarbons — Applied Cases, Reactions & Traps	12 PYQs	Applied Topics
Moderate	Hydrocarbons — Core Principles & Governing Laws	3 PYQs	Core Principle
Standard	Hydrocarbons — Standard Calculations & Problem Archetypes	0 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Hydrocarbons — Applied Cases, Reactions & Traps	$Y^{\wedge}\{ \}$, $\text{Hg}^{\{2+\}}$, $P^{\wedge}\{ \}$, $Q^{\wedge}\{ \}$, $R^{\wedge}\{ \}$
Hydrocarbons — Core Principles & Governing Laws	$3^{\wedge}\{\text{circ}\}$, $60^{\wedge}\{\text{circ}\}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Hydrocarbons — Applied Cases, Reactions & Traps (High Yield — 12 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Hydrocarbons.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "2-methyl propane can be prepared by Wurtz reaction. The haloalkanes taken along with metallic sodium and dry ether are :..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "In the given sequence of reactions, identify ' $P^{\wedge}$ ', ' $Q^{\wedge}$ ', ' $R^{\wedge}$ ' and ' S ' respectively...." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "Sodium ethanoate on heating with soda lime gives ' $X^{\wedge}$ '. Electrolysis of aqueous solution of sodium ethanoate gives ' $Y^{\wedge}$ '. ' $X^{\wedge}$ ' is..." [Ans: N/A]

3.2 Hydrocarbons — Core Principles & Governing Laws (Moderate Yield — 3 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Hydrocarbons.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Hydrocarbons.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Hydrocarbons PYQs.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "Statement I: Staggered conformation of ethane is more stable than the eclipsed conformation. Statement II: The torsional strain in staggered..." **[Ans: N/A]**
- **[2025 Shift] Q#2 (MCQ):** "The major product formed when 1- Bromo -3- Chlorocyclobutane reacts with metallic sodium in dry ether is..." **[Ans: N/A]**
- **[2018 Shift] Q#2 (MCQ):** "When the vapours of tertiary butyl alcohol are passed through heated copper at 573 K , the product formed is..." **[Ans: N/A]**

3.3 Hydrocarbons — Standard Calculations & Problem Archetypes (Standard Yield — 0 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Hydrocarbons.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "Statement I: Staggered conformation of ethane is more stable than the eclipsed conformation. Statement II: The torsional strain in staggered..." **[Ans: N/A]**
- **[2025 Shift] Q#2 (MCQ):** "The major product formed when 1- Bromo -3- Chlorocyclobutane reacts with metallic sodium in dry ether is..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Statement I: Staggered conformation of ethane is more stable than the eclipsed conformation. Statement II: The torsional strain in staggered conformation is more. Read the above statements and choose the correct answer from the options given below.

- (A) Both Statement I and Statement II are false
- (B) Both Statement I and Statement II are true
- (C) Statement I is true but Statement II is false
- (D) Statement I is false but Statement II is true

Correct Answer: None

Solution Analysis: Statement I is true. In ethane, the staggered conformation is more stable because the hydrogen atoms on the two carbon atoms are as far apart as possible. The dihedral angle between C-H bonds is 60° , so the repulsion between bonding electron pairs is minimum. Statement II is false. Torsional strain is not more in the staggered conformation. It is maximum in the eclipsed conformation because the C-H bonds of the two carbon atoms are directly aligned with each other. This causes greater repulsion between bonding electron pairs. So, $\text{Statement I is true but Statement II is false.}$

Benchmark Problem 2: KCET 2024 (Q#1)

2-methyl propane can be prepared by Wurtz reaction. The haloalkanes taken along with metallic sodium and dry ether are :

- (A) chloromethane and 2-chloropropane
- (B) chloroethane and chloromethane
- (C) chloroethane and 1-chloropropane
- (D) chloromethane and 1-chloropropane

Correct Answer: None

Solution Analysis: To prepare 2-methyl propane by Wurtz reaction, the haloalkanes that should be taken along with metallic sodium and dry ether are chloromethane and 2 -chloropropane. The reaction involved is